Lecture 5. Linear Regression

Simple linear regression, loss functions, multiple regression, model selection, interactions

Contents

5.1 Simple linear regression **2**
- 5.1.1 Example: the rail trail . . . 2
- 5.1.2 Prediction and intervals . . . 3
- 5.1.3 Strength of fit and least squares . . . 4
- 5.1.4 A categorical predictor . . . 4

5.2 Loss functions **4**

5.3 Multiple regression **5**
- 5.3.1 Inference for coefficients . . . 6
- 5.3.2 Choosing a model . . . 6

5.4 Other considerations **7**

5.5 Selected exercises **8**

5.1 Simple linear regression

Motivation. Linear regression is the simplest complete answer to how Y depends on predictors; its ideas carry over to later models.

Estimation fits parameters; **prediction** applies the fit to unseen data. y is the *outcome*, x the *predictor*. The model:

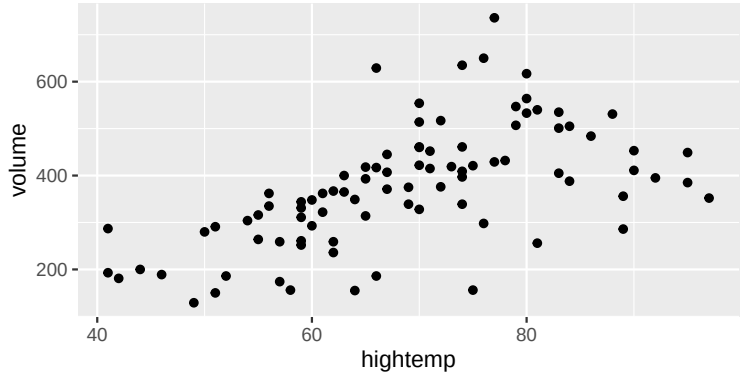
$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad i = 1, \ldots, n,$$

with $\mathbb{E}\epsilon_i = 0$. `lm()` gives $\hat{\beta}_0$, $\hat{\beta}_1$, fitted values $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and residuals $\hat{\epsilon}_i = y_i - \hat{y}_i$.

5.1.1 Example: the rail trail

`mosaicData::RailTrail`: daily `volume` of trail users in Florence, Massachusetts, over 90 days, with temperature, rain, cloud cover and weekday.

```
ggplot(RailTrail, aes(x = hightemp, y = volume)) + geom_point()
```




```
mod <- lm(volume ~ hightemp, data = RailTrail)
tidy(mod)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic       p.value
##   <chr>          <dbl>     <dbl>     <dbl>         <dbl>
## 1 (Intercept)   -17.1     59.4      -0.288 0.774
## 2 hightemp        5.70     0.848     6.72  0.00000000171
```

$\hat{\beta}_1 = 5.70$ users per °F; the intercept (0°F) is outside the data. `summary(mod)` adds residual SE, R^2, F-test:

```
summary(mod)
```

```
##
## Call:
## lm(formula = volume ~ hightemp, data = RailTrail)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -254.562  -57.800    8.737   57.352  314.035
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -17.079     59.395  -0.288    0.774
## hightemp       5.702      0.848   6.724 1.71e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 104.2 on 88 degrees of freedom
## Multiple R-squared:  0.3394, Adjusted R-squared:  0.3319
## F-statistic: 45.21 on 1 and 88 DF,  p-value: 1.705e-09
```

5.1.2 Prediction and intervals

```
predict(mod, newdata = data.frame(hightemp = c(30, 40)))
```

```
##        1        2
## 153.9771 210.9958
```

```
predict(mod, interval = "confidence")[1:3, ]
```

```
##        fit      lwr      upr
## 1 456.1766 423.8301 488.5231
## 2 399.1578 376.2313 422.0843
## 3 404.8597 381.3622 428.3572
```

```
predict(mod, interval = "prediction")[1:3, ]
```

```
## Warning in predict.lm(mod, interval = "prediction"): predictions on current data refer to
    _future_ responses
```

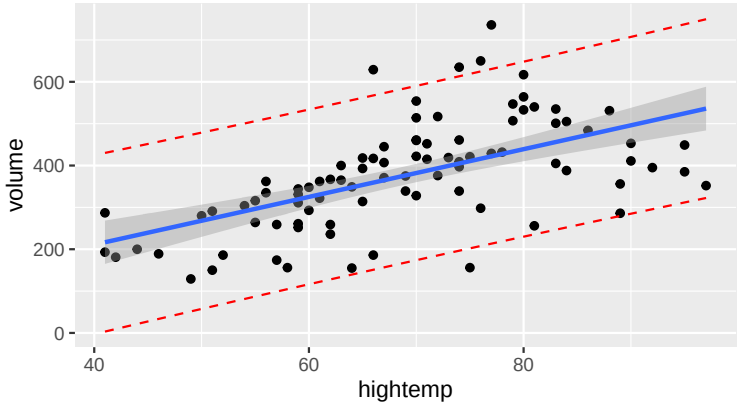
```
##        fit      lwr      upr
## 1 456.1766 246.6175 665.7357
## 2 399.1578 190.8447 607.4709
## 3 404.8597 196.4830 613.2364
```

Confidence interval: mean volume at x; **prediction interval**: one new day, wider by the noise ϵ.

```
pi <- predict(mod, interval = "prediction")
```

```
## Warning in predict.lm(mod, interval = "prediction"): predictions on current data refer to
    _future_ responses
```

```
RailTrail %>%
  mutate(lower.pi = pi[, "lwr"], upper.pi = pi[, "upr"]) %>%
  ggplot(aes(x = hightemp, y = volume)) +
  geom_point() +
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE) +
  geom_line(aes(y = lower.pi), colour = "red", linetype = "dashed") +
  geom_line(aes(y = upper.pi), colour = "red", linetype = "dashed")
```



The two interval formulas [Supplement]

With $S_{xx} = \sum(x_i - \bar{x})^2$ and $s^2 = \text{SSE}/(n-2)$, the intervals at x_0 are

$$\hat{y}_0 \pm t_{n-2,0.975}\, s\sqrt{\frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}} \quad \text{(mean)}, \qquad \hat{y}_0 \pm t_{n-2,0.975}\, s\sqrt{1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}} \quad \text{(new obs.)}.$$

The extra 1 is the new error's variance; both widen away from $\bar{x}$.

5.1.3 Strength of fit and least squares

Beyond the correlation r:

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \in [0,1],$$

the fraction of variation explained relative to the null model $y_i = \beta_0 + \epsilon_i$. Least squares minimises SSE:

$$\sum_i \{y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)\}^2 \le \sum_i \{y_i - (\beta_0 + \beta_1 x_i)\}^2 \quad \text{for all } \beta_0, \beta_1.$$

Least squares is computationally convenient but, like the mean, outlier-sensitive. Generally, *a statistical method optimises a goodness-of-fit measure over a family of models*, by formula or by algorithm.

5.1.4 A categorical predictor

With the indicator `weekday`, $\text{volume}_i = \beta_0 + \beta_1 \text{weekday}_i + \epsilon_i$ fits the two group means:

```
RailTrail %>% group_by(weekday) %>% summarise(mean_volume = mean(volume))
```

```
## # A tibble: 2 x 2
##   weekday mean_volume
##   <lgl>         <dbl>
## 1 FALSE          431.
## 2 TRUE           350.
```

```
coef(lm(volume ~ weekday, data = RailTrail))
```

```
## (Intercept) weekdayTRUE
##   430.71429   -80.29493
```

About 80 fewer riders on weekdays. The alphabetically first level is the **reference group**; recoding flips the sign:

```
RailTrail <- RailTrail %>%
  mutate(day = ifelse(weekday == 1, "weekday", "weekend/holiday"))
coef(lm(volume ~ day, data = RailTrail))
```

```
##        (Intercept) dayweekend/holiday
##          350.41935           80.29493
```

5.2 Loss functions

Motivation. The loss function defines "best fit"; different losses differ most with outliers.

A **loss function** $L(y, \hat{y})$: squared (L_2) $(y-\hat{y})^2$ or absolute (L_1) $|y-\hat{y}|$; L_2 penalises large residuals more. Fitting minimises the **empirical risk**

$$\hat{R}(\beta_0, \beta_1) = \frac{1}{n}\sum_i L(y_i, \hat{y}_i),$$

(MSE for L_2, MAE for L_1). Least squares minimises the L_2 risk; zero derivatives give the estimating equations $\frac{1}{n}\sum_i (y_i - \hat{y}_i) = 0$ and $\frac{1}{n}\sum_i (y_i - \hat{y}_i)x_i = 0$, whose solution is

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

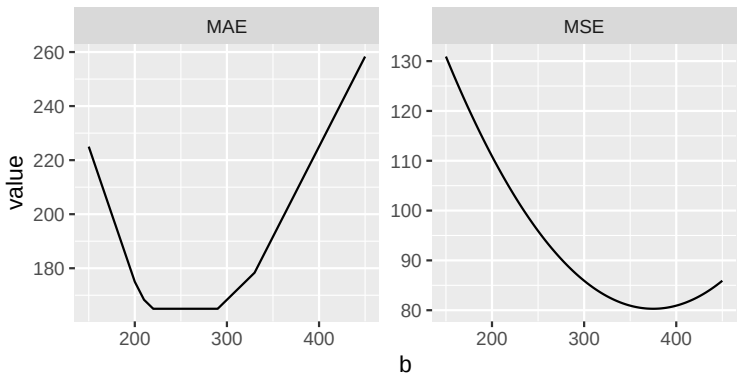
Proposition 5.1 (constant model)

For the constant model $y = \beta_0 + \epsilon$, the minimiser of the L_2 risk is $\bar{y}$ and a minimiser of the L_1 risk is any median of $y_1, \ldots, y_n$.

Proof. L_2: $\frac{d}{d\beta_0}\frac{1}{n}\sum(y_i - \beta_0)^2 = -\frac{2}{n}\sum(y_i - \beta_0) = 0$ gives $\beta_0 = \bar{y}$, and the risk is a convex parabola. L_1: away from the data points, $\frac{d}{d\beta_0}\frac{1}{n}\sum|y_i - \beta_0| = \frac{1}{n}(\#\{y_i < \beta_0\} - \#\{y_i > \beta_0\})$. The risk is convex and piecewise linear, decreasing while fewer than half the points lie below β_0 and increasing once more than half do, so it is minimised at a median. ■

Example (bubble-tea sales). Sales 200, 210, 220, 290, 330: mean 250 (L_2), median 220 (L_1). Adding 1000 moves the mean to 375, the median to 255; with six points any value in $[220, 290]$ minimises the MAE.

```
y <- c(200, 210, 220, 290, 330, 1000)
tibble(b = seq(150, 450, by = 1)) %>%
  mutate(MSE = map_dbl(b, ~ mean((y - .x)^2)) / 1000,
         MAE = map_dbl(b, ~ mean(abs(y - .x)))) %>%
  pivot_longer(-b) %>%
  ggplot(aes(b, value)) + geom_line() + facet_wrap(~ name, scales = "free_y")
```



MSE (in thousands): smooth, unique minimum, outlier-sensitive. MAE: piecewise linear, flat on $[220, 290]$, robust.

5.3 Multiple regression

Motivation. Multiple regression estimates each effect holding the others fixed.

$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon$; β_j is the change in y per unit of x_j *holding the others fixed.*

```
mod_plane <- lm(volume ~ hightemp + precip, data = RailTrail)
tidy(mod_plane)
```

```
## # A tibble: 3 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)   -31.5     55.2      -0.571 5.70e- 1
## 2 hightemp        6.12     0.794     7.70  1.97e-11
## 3 precip       -153.      39.3      -3.90  1.90e- 4
```

An inch of rain lowers ridership by about 153 at fixed temperature.

5.3.1 Inference for coefficients

Normal-approximation 95% interval: $\hat{\beta}_j \pm 1.96\,\mathrm{SE}(\hat{\beta}_j)$; `confint()` uses t quantiles:

```
mod_p_planes <- lm(volume ~ hightemp + precip + weekday, data = RailTrail)
tidy(mod_p_planes)
```

```
## # A tibble: 4 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)    19.3     60.3       0.320 7.50e- 1
## 2 hightemp        5.80     0.799     7.26  1.59e-10
## 3 precip       -146.      38.9      -3.74  3.27e- 4
## 4 weekdayTRUE   -43.1     22.2      -1.94  5.52e- 2
```

```
confint(mod_p_planes)
```

```
##                   2.5 %      97.5 %
## (Intercept) -100.630740 139.2683959
## hightemp       4.213254   7.3880945
## precip      -222.927105 -68.2908651
## weekdayTRUE  -87.265016   0.9763902
```

Bootstrap (Lecture 4): resample days, refit, take the sd of the coefficients.

```
set.seed(1)
n <- nrow(RailTrail)
coef_fun <- function(data) lm(volume ~ hightemp + precip + weekday, data = data)$coef
boot <- 1:500 %>%
  map_dfr(~ RailTrail %>% slice_sample(n = n, replace = TRUE) %>% coef_fun())
round(coef(mod_p_planes)["hightemp"] + c(-1.96, 1.96) * sd(boot$hightemp), 2)
```

```
## [1] 4.2 7.4
```

Close to the model-based interval.

5.3.2 Choosing a model

R^2 always grows with more variables; **Occam's razor** prefers the simpler model. Candidates (`volume ~ .` uses all):

```
mod1 <- lm(volume ~ hightemp + weekday, data = RailTrail)
mod2 <- lm(volume ~ hightemp + cloudcover + weekday, data = RailTrail)
mod3 <- lm(volume ~ . - day, data = RailTrail)
map_dfr(list(mod1 = mod1, mod2 = mod2, mod3 = mod3), glance, .id = "model") %>%
  select(model, r.squared, adj.r.squared, AIC, BIC)
```

```
## # A tibble: 3 x 5
##   model r.squared adj.r.squared   AIC   BIC
##   <chr>     <dbl>         <dbl> <dbl> <dbl>
## 1 mod1      0.374         0.359 1093. 1103.
## 2 mod2      0.459         0.440 1082. 1094.
```

```
## 3 mod3      0.511         0.469 1081. 1103.
```

BIC prefers `mod2`; adjusted R^2 and AIC slightly favour the full model. Criteria penalising size:

- adjusted $R^2_{\text{adj}} = 1 - (1 - R^2)\dfrac{n-1}{n-p-1}$, with p the number of predictors;
- AIC (Akaike's information criterion) and BIC (Bayesian, with a stronger penalty on p), smaller is better;
- Mallows' C_p.

All-subset regression searches 2^p models; **stepwise regression** goes backward, forward or both (`MASS::stepAIC()`); **penalized regression** (Lecture 7) shrinks coefficients. In-sample selection can **overfit**; cross-validation (Lecture 6) guards against it, essential for flexible methods.

5.4 Other considerations

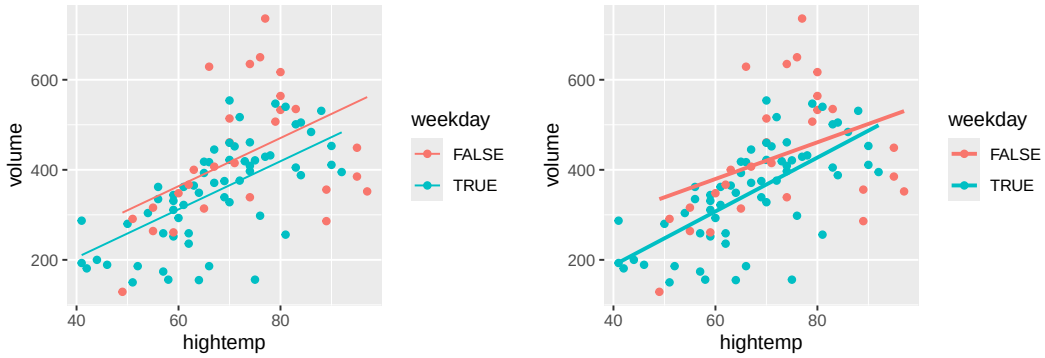
Factor variables are coded by dummy (indicator) variables: *reference coding* drops one level, *one-hot encoding* keeps all (and then drops the intercept). Ordered categories may be treated as numeric scores.

Interactions. The linear model is *additive* and *linear*. With 0/1 x_2 (weekday), $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$ gives parallel lines (intercepts $\hat{\beta}_0$, $\hat{\beta}_0 + \hat{\beta}_2$). Adding $x_1 x_2$,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_1 x_2,$$

gives slope $\hat{\beta}_1$ on weekends and $\hat{\beta}_1 + \hat{\beta}_3$ on weekdays: an **interaction model**.

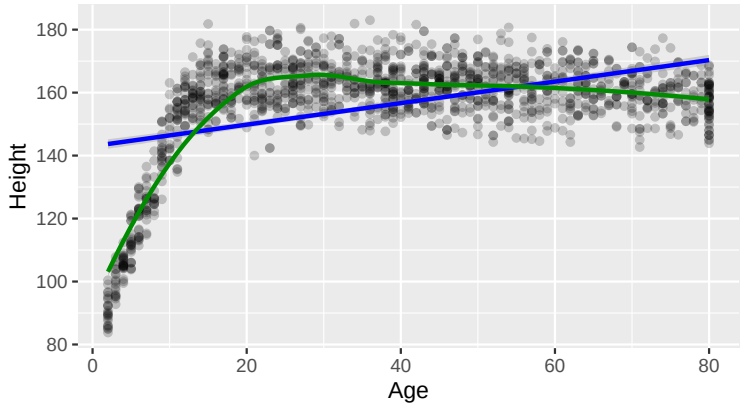
```
ggplot(RailTrail, aes(hightemp, volume, colour = weekday)) +
  geom_point() +
  geom_line(aes(y = fitted(lm(volume ~ hightemp + weekday, data = RailTrail))))
ggplot(RailTrail, aes(hightemp, volume, colour = weekday)) +
  geom_point() + geom_smooth(method = "lm", formula = y ~ x, se = FALSE)
```



Left: parallel slopes; right: separate slopes (interaction).

Nonlinear relationships. Female height against age (`NHANES`): a line misfits, LOESS follows the growth curve with a wider band (Lecture 10).

```
NHANES %>% filter(Gender == "female", !is.na(Height)) %>%
  slice_sample(n = 1500) %>%
  ggplot(aes(Age, Height)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm", formula = y ~ x, colour = "blue") +
  geom_smooth(method = "loess", formula = y ~ x, colour = "green4")
```



5.5 Selected exercises

Motivation. Least-squares facts used without proof, each from the estimating equations.

Exercise 5.1 (R^2 in simple regression)

For the least-squares line with residuals $e_i = y_i - \hat{y}_i$, and assuming SST > 0:
(a) show that $\sum_i e_i = 0$, $\sum_i x_i e_i = 0$ and SST $=$ SSR $+$ SSE with SSR $= \sum_i (\hat{y}_i - \bar{y})^2$; conclude $0 \le R^2 \le 1$;
(b) show that $R^2 = r^2$, the squared sample correlation of x and y.

Solution [Supplement].

(a) The first two identities are the estimating equations of Section 5.2, multiplied by n. Since $\hat{y}_i - \bar{y} = \hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y}$ is a linear combination of 1 and x_i, they give $\sum_i (\hat{y}_i - \bar{y}) e_i = 0$. Expanding $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + e_i$,

$$\text{SST} = \text{SSR} + \text{SSE} + 2\sum_i (\hat{y}_i - \bar{y}) e_i = \text{SSR} + \text{SSE}.$$

Both terms are ≥ 0, so $R^2 = 1 - \text{SSE/SST} = \text{SSR/SST} \in [0, 1]$.

(b) Write $S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y})$ and $S_{yy} = \text{SST}$. From $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ we get $\hat{y}_i - \bar{y} = \hat{\beta}_1 (x_i - \bar{x})$, so

$$\text{SSR} = \hat{\beta}_1^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}}, \qquad R^2 = \frac{S_{xy}^2}{S_{xx} S_{yy}} = r^2.$$

■

Exercise 5.2 (indicators reproduce group means)

(a) For $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with $x_i \in \{0, 1\}$ and both groups non-empty, show that $\hat{\beta}_0 = \bar{y}_0$ and $\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$, the group means and their difference.
(b) For the interaction model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon$ with $x_2 \in \{0, 1\}$, show that the least-squares fit equals the two simple regressions of y on x_1 fitted separately in the groups $x_2 = 0$ and $x_2 = 1$ (assume x_1 is not constant in either group).

Solution [Supplement].

(a) Put $a = \beta_0$ and $b = \beta_0 + \beta_1$; the map $(\beta_0, \beta_1) \mapsto (a, b)$ is a bijection. The residual sum of squares splits as

$$\sum_{i: x_i = 0} (y_i - a)^2 + \sum_{i: x_i = 1} (y_i - b)^2,$$

each minimised separately; by Proposition 5.1 the minimisers are $\hat{a} = \bar{y}_0$ and $\hat{b} = \bar{y}_1$, hence $\hat{\beta}_0 = \bar{y}_0$ and $\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$.

(b) Put $(a_0, a_1) = (\beta_0, \beta_1)$ and $(b_0, b_1) = (\beta_0 + \beta_2, \beta_1 + \beta_3)$, again a bijection. The fitted value is $a_0 + a_1 x_1$ when $x_2 = 0$ and $b_0 + b_1 x_1$ when $x_2 = 1$, so

$$\text{RSS} = \sum_{i: x_{i2}=0} (y_i - a_0 - a_1 x_{i1})^2 + \sum_{i: x_{i2}=1} (y_i - b_0 - b_1 x_{i1})^2.$$

Each sum is a within-group simple-regression RSS in its own parameters, minimised by the separate least-squares lines (unique since x_1 varies). The interaction model adds a single model in which β_3, the slope difference, can be tested. ■

Exercise 5.3 (precision of the slope)

Treat $x_1, \dots, x_n$ as fixed and let the errors be uncorrelated with mean 0 and variance σ^2.
(a) Show that $\hat{\beta}_1$ is unbiased with $\text{Var}(\hat{\beta}_1) = \sigma^2 / S_{xx}$.
(b) If the x_i must lie in $[L, U]$, show that $S_{xx} \le n(U-L)^2/4$, with equality when half of the points (for even n) are at each end. What does this say about designing a study?

Solution [Supplement].

(a) Because $\sum_i (x_i - \bar{x}) = 0$, $\hat{\beta}_1 = \sum_i (x_i - \bar{x}) y_i / S_{xx}$, a linear combination of the y_i with weights $w_i = (x_i - \bar{x})/S_{xx}$. Then

$$\mathbb{E}\hat{\beta}_1 = \sum_i w_i (\beta_0 + \beta_1 x_i) = \beta_1 \frac{\sum_i (x_i - \bar{x}) x_i}{S_{xx}} = \beta_1,$$

since $\sum_i w_i = 0$ and $\sum_i (x_i - \bar{x}) x_i = S_{xx}$. With uncorrelated errors, $\text{Var}\,\hat{\beta}_1 = \sigma^2 \sum_i w_i^2 = \sigma^2 / S_{xx}$.

(b) Let $m = (L + U)/2$. The mean minimises the sum of squared deviations (Proposition 5.1), so $S_{xx} \le \sum_i (x_i - m)^2 \le n\{(U-L)/2\}^2$, because $|x_i - m| \le (U-L)/2$. With half the points at each end, $\bar{x} = m$ and $|x_i - m| = (U-L)/2$, so equality holds. Spreading x maximises precision, but some interior points are needed to check linearity. ■